

BankAccountManagementSystem

Java Programming Laboratory Experiment

Experiment No.	03
Date	11/09/2026
Title	Bank Account Management System using Classes and Objects

Objective / Aim

To design and implement a Bank Account Management System in Java using classes and objects, demonstrating data members, object creation, methods, conditional statements, deposits, withdrawals, and displaying account details.

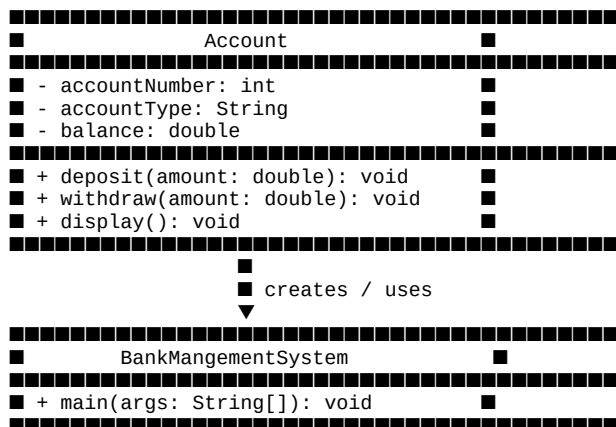
System Requirements

HARDWARE	SOFTWARE
• Standard PC/Laptop • Intel Core i3/i5 processor or equivalent • Minimum 4 GB RAM • Minimum 500 MB free storage	• Operating System: Windows / Linux / macOS • Java Development Kit (JDK 17 or JDK 21) • Text Editor / IDE: Eclipse, IntelliJ IDEA, NetBeans, or VS Code • Command Prompt / Terminal

Algorithm / Logical Description

1. Create an Account class.
2. Declare account number, account type, and balance as data members.
3. Define deposit() to add money to the account balance.
4. Define withdraw() to check available balance and deduct the withdrawal amount when sufficient balance is available.
5. Define display() to show account details.
6. Create an Account object in main().
7. Assign account number, account type, and initial balance.
8. Display the initial account details.
9. Perform deposit and withdrawal transactions.
10. Display the updated account details after transactions.

UML Class Diagram



Relationship: The BankMangementSystem class creates and uses an object of the Account class. The Account class contains data members and methods for managing account transactions.

Source Code

```
// Account class represents a bank account
class Account {
// Account class represents a general bank account
class Account {
    // Data members of the Account class
    int accountNumber;
    double balance;
    String accountType;
```

```

// Parameterized constructor to initialize account details
Account(int accountNumber, double balance, String accountType) {
    this.accountNumber = accountNumber;
    this.balance = balance;
    this.accountType = accountType;
}

// Method to deposit money into the account
void deposit(double amount) {
    // Check whether the deposit amount is positive
    if (amount > 0)
        balance += amount;
}

// Method to withdraw money from the account
void withdraw(double amount) {
    // Check whether the amount is valid and balance is sufficient
    if (amount > 0 && amount <= balance)
        balance -= amount;
    else
        System.out.println("Insufficient balance");
}

// Method to calculate interest
// Default implementation for a general account
void calculateInterest() {
    System.out.println("No interest");
}

// Method to display account details
void display() {
    // Display account number
    System.out.println("Account Number: " + accountNumber);

    // Display account type
    System.out.println("Account type: " + accountType);

    // Display current balance
    System.out.println("Balance: " + balance);
}

// SavingsAccount class inherits Account
class SavingsAccount extends Account {
    // Interest rate for savings account
    private double interestRate;

    // Constructor to initialize savings account
    SavingsAccount(int no, double balance, double rate) {
        // Calling parent class constructor
        super(no, balance, "Savings Account");

        // Assigning interest rate
        interestRate = rate;
    }

    // Overriding calculateInterest() method
    @Override
    void calculateInterest() {
        // Calculating interest based on current balance
        double interest = balance * interestRate / 100;

        // Adding interest to the account balance
        balance += interest;

        // Displaying calculated interest
        System.out.println("Interest: " + interest);
    }
}

// CurrentAccount class inherits Account
class CurrentAccount extends Account {
    // Maximum overdraft amount allowed
    private double overdraftLimit;

    // Constructor to initialize current account
    CurrentAccount(int no, double balance, double limit) {
        // Calling parent class constructor
        super(no, balance, "Current Account");
    }
}

```

```

        // Assigning overdraft limit
        overdraftLimit = limit;
    }

    // Overriding withdraw() method to support overdraft
    @Override
    void withdraw(double amount) {

        // Check whether withdrawal is within overdraft limit
        if (amount > 0 && amount <= balance + overdraftLimit)
            balance -= amount;
        else
            System.out.println("Exceeds overdraft limit");
    }
}

// Main class of the Bank Account Management System
public class BankAccountManagement {

    // Static method to transfer money between two accounts
    static void transfer(Account from, Account to, double amount) {

        // Check whether transfer amount is valid
        // and sufficient balance is available
        if (amount > 0 && amount <= from.balance) {

            // Deduct amount from source account
            from.balance -= amount;

            // Add amount to destination account
            to.balance += amount;

            // Display successful transfer message
            System.out.println("Transfer successful");

        } else {

            // Display message when transfer fails
            System.out.println("Transfer failed");
        }
    }

    // Main method - program execution starts here
    public static void main(String[] args) {

        // Creating a SavingsAccount object
        SavingsAccount savings =
            new SavingsAccount(2002, 500000, 3);

        // Creating a CurrentAccount object
        CurrentAccount current =
            new CurrentAccount(2003, 5000, 3000);

        // Displaying initial savings account details
        savings.display();

        // Depositing Rs.3000 into savings account
        savings.deposit(3000);

        // Calculating and adding interest
        savings.calculateInterest();

        // Displaying initial current account details
        current.display();

        // Withdrawing Rs.6000 using overdraft facility
        current.withdraw(6000);

        // Transferring Rs.2000 from savings to current account
        transfer(savings, current, 2000);

        // Displaying savings account details after transactions
        savings.display();

        // Displaying current account details after transactions
        current.display();
    }
}

```

Completion and Execution Commands

Save the source code exactly as:

```
BankAccountManagement.java
```

Compile the program:

```
javac BankAccountManagement.java
```

Run the program:

```
java BankAccountManagement
```

Output Screen

Account Number: 2002

Account type: Savings Account

Balance: 500000.0

Interest: 15090.0

Account Number: 2003

Account type: Current Account

Balance: -1000.0

Transfer successful

Account Number: 2002

Account type: Savings Account

Balance: 513090.0

Account Number: 2003

Account type: Current Account

Balance: 1000.0

Results

The Bank Account Management System was successfully implemented in Java using classes and objects. The program successfully performs deposit, withdrawal, interest calculation, overdraft, and transfer operations and displays account details.